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ORIGINAL PAPER

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The auxin-responsive *GH3* gene family in rice (*Oryza sativa*)

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Abstract Auxin regulates plant growth and development by altering the expression of diverse genes. Among these, the genes of *Aux/IAA*, *SAUR*, and *GH3* classes have been extensively studied in dicots, but little information is available on monocots. We have identified 12 members of *GH3* gene family in rice using sequences of full-length cDNA clones available from KOME and analysis of the whole genome sequence of rice. The genomic organization as well as chromosomal location of all the *OsGH3* genes is reported. The rice GH3 proteins can be classified in two groups (groups I and II) on the basis of their phylogenetic relationship with *Arabidopsis* GH3 proteins. Based upon the sequences available in the database, not a single group III GH3 protein could be identified in rice. An extensive survey of EST sequences of other monocots led to the conclusion that although *GH3* gene family is highly conserved in both dicots and monocots but the group III is conspicuous by its absence in monocots. The in silico analysis has been complemented with experimental data to quantify transcript levels of all *GH3* gene family members. Using real-time polymerase chain reaction, the organ-specific expression of individual *OsGH3* genes in light- and dark-grown seedlings/plants has been examined. The transcript abundance of nearly all *OsGH3* genes is enhanced on auxin treatment, with the effect more pronounced on *OsGH3-1*, *-2*, and *-4*. The functional validation of these genes in transgenics or analysis of gene-specific insertional mutants will help in elucidating their precise role in auxin signal transduction.

Keywords Auxin · *GH3* gene family · Phylogenetic analysis · Rice

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Introduction

The phytohormone auxin plays a critical role in regulating a number of plant responses including cell elongation, cell division, phototropism, gravitropism, root initiation, and apical dominance. In an attempt to understand the molecular mechanism of auxin action, a number of early or primary auxin-responsive genes have been identified and characterized from different plant species including pea, soybean, *Arabidopsis*, tobacco, mung bean, and cucumber (Hagen and Guilfoyle 2002). These auxin-responsive genes have been broadly grouped into three major classes: *Aux/IAA*, *SAUR*, and *GH3* (Guilfoyle 1999). A number of putative auxin-responsive elements (AuxREs) have been defined within upstream promoter regions of auxin-responsive genes, such as one or more copies of a conserved motif, TGTCTC, or some variation of this motif (e.g., TGTCCC, TGTCAC), which confer auxin responsiveness (Hagen et al. 1991; Li et al. 1994; Liu et al. 1994; Ulmasov et al. 1995; Guilfoyle 1999). The DNA-binding domain of auxin-responsive factors (ARFs) binds to AuxREs and regulates the expression of auxin-responsive genes (Kim et al. 1997; Ulmasov et al. 1997; Tiwari et al. 2003).

Initially, early auxin-responsive genes of *Aux/IAA* class were isolated from soybean (Walker and Key 1982; Ainley et al. 1988) and later from pea (Theologis et al. 1985), mung bean (Yamamoto et al. 1992), *Arabidopsis* (Conner et al. 1990; Abel et al. 1995), and rice (Thakur et al. 2001). The *SAUR* genes have also been isolated and studied in plants such as soybean (Hagen et al. 1984), mung bean (Yamamoto et al. 1992), and *Arabidopsis* (Gil et al. 1994). The first *GH3* gene was isolated from soybean by differential screening as an early auxin-responsive gene (Hagen et al. 1984). The soybean *GH3* transcription was shown to be specifically induced by biologically active auxins within 5 min of exogenous auxin application. Also, unlike *SAUR*s and *Aux/IAA*s, *GH3* mRNAs do not accumulate in response to protein synthesis inhibitor cycloheximide (Franco et al. 1990).

The *GH3* homologues have been found in other plants too (Guilfoyle et al. 1993; Roux and Perrot-Rechenmann 1997). The *GH3* genes are represented as a multigene family in the *Arabidopsis* genome comprising of 20 members (Hagen and Guilfoyle 2002). Many of the *AtGH3* genes were induced by exogenous application of auxin (Hagen and Guilfoyle 2002; Tanaka et al. 2002). Light also regulates the transcript levels of *Arabidopsis GH3* genes through phytochromes A and B (Nakazawa et al. 2001; Tepperman et al. 2001; Tanaka et al. 2002; Takase et al. 2003, 2004). The analysis of an *Arabidopsis* mutant identified FIN219, a GH3 protein, as a phytochrome A signaling component having a crucial role in photomorphogenesis (Hsieh et al. 2000); it negatively regulates COP1, a key repressor of photomorphogenic development. Two other *Arabidopsis* GH3 proteins, DFL1 and YDK1, also identified by mutant analysis, were shown to negatively regulate shoot and hypocotyl cell elongation and lateral root formation (Nakazawa et al. 2001; Takase et al. 2004). *Arabidopsis* GH3 proteins have been reported to adenylate plant hormones such as indole-3-acetic acid (IAA), jasmonic acid (JA), and salicylic acid (SA) (Staswick et al. 2002). Recently, some of the *Arabidopsis GH3* genes have been shown to encode IAA-amido synthetases, which synthesize a diversity of IAA–amino acid (aa) conjugates to help maintain auxin homeostasis (Staswick et al. 2005).

To our knowledge, there is no report on the characterization of any *GH3* gene from a monocot species, although several EST/cDNA sequences are available in the database. Here, we report the identification and a comprehensive analysis of the *GH3* gene family from rice (*Oryza sativa*), the model monocot plant. The work involved the identification of *GH3* gene family from rice, analysis of their genomic organization, chromosomal distribution, sequence homology, and phylogenetic relationship. Systematic identification of ESTs representing different members of the *GH3* gene family, along with real-time PCR analysis, indicated their differential expression in various organs/tissues and in response to light and auxin treatment. Extensive survey of ESTs of this family has also been carried out in other monocots including wheat, maize, sorghum, sugarcane, and barley.

Materials and methods

Identification of *GH3* gene family sequences

The predicted longest open-reading frames (ORFs) from the Knowledge-Based Oryza Molecular Biological Encyclopedia (KOME, http://www.cdna01.dna.affrc.go.jp/cDNA) (Kikuchi et al. 2003), National Centre for Biotechnology Information (NCBI, http://www.ncbi.nlm.nih.gov/BLAST), and TIGR genomic and annotated database resources (http://www.tigrblast.tigr.org/euk-blast) were used for the identification of *GH3* homologues in rice. The *Arabidopsis* GH3 protein and cDNA sequences were downloaded from The *Arabidopsis* Information Resource (TAIR) (Garcia-Hernandez et al. 2002). The protein sequences of 20 GH3 proteins from *Arabidopsis* were used to search for their homologues in rice in KOME [28,469 full-length cDNA (FL-cDNA) clones] using TBLASTN program (Altschul et al. 1997). This led to the identification of 12 predicted GH3 proteins, which included nine nonredundant clones having high-sequence similarity with AtGH3 proteins. Additional members of rice *GH3* gene family were identified by TBLASTN search in TIGR database. The predicted *GH3* genes in *O. sativa* were named *OsGH3* (Table 1).

Sequence analysis

The number and positions of exons and introns for individual *OsGH3* genes were determined by comparison of the cDNAs with their corresponding genomic DNA sequences. The position of each gene on rice chromosomes was found by BLASTN search in genomic sequences of rice chromosome pseudomolecules available at TIGR (Release 3). Multiple-sequence alignments were done using the ClustalX (version 1.83) program (Thompson et al. 1997) and the phylogenetic analysis carried out by neighbor-joining method (Saitou and Nei 1987). The unrooted phylogenetic tree was displayed using the NJPLOT program (Perriere and Gouy 1996). The DNA and protein sequence analysis were performed using Gene Runner program version 3.04. Pairwise comparisons were done with the DNASTAR MegAlign 4.03 package to determine the sequence identities. Conserved residues were identified by alignment using ClustalX.

Plant material and growth conditions

Rice (*O. sativa* L. spp. *indica* var. Pusa Basmati 1) seeds were obtained from Regional Station of the Indian Agricultural Research Institute, Karnal. After disinfection with 0.1% $HgCl_2$ for 1 h and thorough washing with reverse-osmosis (RO) water, seeds were soaked overnight in RO water. Etiolated shoots, roots, and green shoots were harvested from 6-day-old seedlings grown on cotton saturated with RO water, at 28±1°C, either in complete darkness or in a culture room with a daily photoperiodic cycle of 14 h light and 10 h dark. Flowers were collected from field-grown rice plants. Calli were raised as previously described (Jain et al. 2004). For auxin treatment, coleoptile segments (10 mm) from the apical portion of 3-day-old etiolated rice seedlings were incubated in KPSC buffer (10 mM potassium phosphate, pH 6.0, 2% sucrose, 50 μM chloramphenicol) for 16 h to deplete endogenous auxins, with the buffer changed every 2 h, and then transferred to fresh buffer with 30 μM concentration of 2,4-dichlorophenoxyacetic acid (2,4-D) for 3 h.

Table 1 *GH3* gene family in rice

Gene name[a]	Accession no.[b]	Protein length[c] (aa)	ORF length[d] (bp)	No. of introns[e]	Genomic locus[f]			Nearest marker[g]	EST[h]	
					BAC/PAC name	Accession no.	cM position		Accession no.	Frequency
OsGH3-1	AK063368	610	1833	2	B1070A12	AP003406	136.1	E30358S	NF	0
OsGH3-2	LOC_Os01g55940*	614	1845	2	P0403C05	AP003239	132	R37	AU068252	8
OsGH3-3	AK072125	462	1389	2	P0483F08	AP002094	32.4	C52458S	CB660383	2
OsGH3-4	AK101932	629	1890	1	OSJNBa0017K09	AC130597	103.1	R3802S	BI801112	2
OsGH3-5	AK071721	581	1746	3	OSJNBa0009C07	AC137608	120.6	S11036	CB617880	28
OsGH3-6	AK106538	488	1467	0	OJ1607_F09	AC104283	22.5–24.7	E60961	NF	0
OsGH3-7	AK099376	620	1863	3	P0596H06	AP003620	65.8	E769S	CA767214	7
OsGH3-8	AK101193	605	1818	1	OJ1710_H11	AP003846	83.3–84.1	R1245	AU056581	4
OsGH3-9	AK106839	441	1326	0	OJ1065_B06	AP003804	81.4	E1186S	NF	0
OsGH3-10	LOC_Os07g38860*	478	1437	2	OJ1065_B06	AP003804	81.4	E1186S	NF	0
OsGH3-11	LOC_Os07g47490*	633	1902	4	P0470D12	AP004300	114.5	C53986	CA767416	3
OsGH3-12	LOC_Os11g08340*	613	1842	2	OSJNBb0009F15	AC128644	20.3–27.8	S10616A	NF	0

[a]Systematic designation given to rice *GH3* genes
[b]Accession no. of full-length cDNA sequence from KOME (http://cdna.01.dna.affrc.go.jp/cDNA/) or TIGR locus ID (marked with asterisk) in release 3
[c]Length [number of amino acids (aa)] of the deduced polypeptide
[d]Length of open-reading frame in base pairs
[e]Number of introns present within ORF
[f]Name, accession number, and approximate cM position of the BAC/PAC clone in which *OsGH3* gene is present
[g]Nearest marker to the *OsGH3* gene
[h]Representative EST accession no. and number of ESTs present in Genbank release 111204.NF (not found)

RNA isolation and quantitative real-time PCR expression analysis

Total RNA was extracted using the RNeasy Plant mini kit (Qiagen, Germany) according to the manufacturer's instructions, followed by DNase I treatment to remove any genomic DNA contamination. For each RNA sample, absorption at 260 nm was measured and RNA concentration calculated as A_{260}×40 (μg/ml)×dilution factor.

First strand cDNA was synthesized by reverse transcribing 1.5 μg of total RNA (reaction volume 50 μl) using high-capacity cDNA Archive kit (Applied Biosystems, USA) according to the manufacturer's instructions. Primers were designed by using the primer design software Primer Express 2.0 (PE Applied Biosystems, USA). To ensure that the primers amplify a unique and desired cDNA segment, each pair of primers was checked in the BLAST program in rice genomic sequence available in TIGR database. The primer sequences are listed in Table S1 of Supplementary material. The diluted cDNA samples were used as template and mixed with 200 nM of each primer and SYBR Green PCR Master Mix (Applied Biosystems, USA) for real-time PCR analysis, using ABI Prism 7000 Sequence Detection System and Software (PE Applied Biosystems, USA). PCR reactions were performed using the following parameters: 2 min at 50°C, 10 min at 95°C, 40 cycles of 15 s at 95°C, and 1 min at 60°C in 96-well optical reaction plates (Applied Biosystems, USA). The identities of the amplicons and the specificity of the reaction were verified by agarose gel electrophoresis and melting curve analysis, respectively. The relative mRNA levels of the individual *OsGH3* genes in different RNA samples were computed with respect to the internal standard, *UBQ5*, to normalize for variance in the quality of RNA and the amount of input cDNA. At least two different RNA isolations and cDNA syntheses were used for quantification, and each cDNA was measured in triplicate.

EST analysis

The search for ESTs in rice and other monocots was performed by comparing the rice *GH3* cDNA sequences against EST sequences available at NCBI in BLASTN searches. The *OsGH3* cDNA sequences were searched using MEGABLAST tool against the EST databases of rice (*O. sativa*), wheat (*Triticum aestivum*), corn (*Zea mays*), *Sorghum bicolor*, sugarcane (*Saccharum officinarum*), and barley (*Hordeum vulgare*) at NCBI (http://www.ncbi.nlm.nih.gov/dbEST/index.html). The searches were limited to one species at a time to detect all the existing ESTs that showed significant similarity. Details of only top hits of the BLASTN search results for each *OsGH3* gene showing a bit score of 500 or more were extracted into a separate file. An optimized value of a word size of 11 and expect value of 10, without the low-complexity filter, were used for analysis in order to record the optimum number of matches between rice genes and ESTs from other monocots.

Results and discussion

GH3 gene family in rice

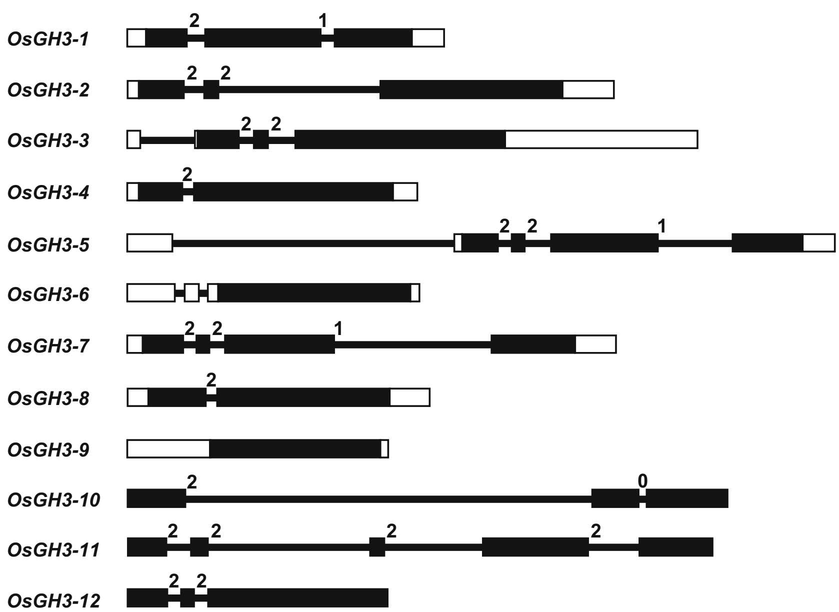
The information on rice genomic sequence (Goff et al. 2002; Yu et al. 2002; http://www.tigr.org/tdb/e2k1/osa1/; http://www.rgp.dna.affrc.go.jp/) provides a powerful tool to identify putative homologous proteins by database searches with genes of known function from other organisms. In an attempt to identify GH3 protein-coding genes in rice, a TBLASTN search of FL-cDNA clones of rice available at KOME was performed using 20 AtGH3 proteins as query, which identified nine nonredundant clones (among a total of 12) and were designated as *O. sativa GH3* genes (*OsGH3-1* to *-9*; Table 1). To identify additional members of this family not represented in KOME cDNA collection, the whole rice genome, dynamically translated in all reading frames, was analyzed employing TBLASTN search, using the 20 GH3 proteins of *Arabidopsis* and nine nonredundant rice GH3 proteins identified through KOME. Three additional members, *OsGH3-10* to *-12*, showing significant similarity with other OsGH3 proteins could be identified in this search. The overall analysis of the complete genome of rice revealed that auxin-inducible *GH3* gene family is comprised of 12 members. However, in *A. thaliana*, 19 *GH3* genes, along with an additional partial gene (coding only for amino-terminal third of the protein), have been reported (Hagen and Guilfoyle 2002).

A comparative study was done for all the OsGH3 proteins found in KOME with the corresponding annotated proteins present in the NCBI and TIGR databases (Supplementary material, Table S2). The amino acid sequences of all were found to be essentially identical in three databases, except for OsGH3-2, -3, and -6. Therefore, the amino acid and cDNA sequences of *OsGH3-2*, *-3*, and *-6* were examined in detail. The C-terminal amino acid sequence of OsGH3-2 protein from KOME did not show any homology with other known GH3 proteins. Its cDNA sequence was aligned with annotated cDNA sequence from NCBI and TIGR and the corresponding genomic sequence [phage artificial chromosome (PAC) clone P0403C05]. Two nucleotides (GC) were found to be missing in KOME cDNA sequence within ORF leading to the change in coding frame. However, the corresponding protein from TIGR and NCBI were identical and showed significant similarity with other GH3 proteins and included in Table 1 for further analysis. The OsGH3-3 protein predicted by KOME and TIGR was identical (462 aa). However, NCBI predicted it to be 623 aa long, the C-terminal region of which did not show identity with the sequence available in KOME or TIGR database. The cDNA sequence of *OsGH3-3* was supported by two full-length cDNAs from KOME (AK072125 and AK120515) and the two ESTs (CB660383 and CB660384); thus, it is most likely that OsGH3 protein comprises 462 aa (Table 1). The OsGH3-6 protein was predicted to be 488, 486, and 908 amino acid long in KOME, NCBI, and TIGR databases, respectively; the NCBI-annotated sequence showed 100% identity with KOME OsGH3-6 sequence except that two amino acids were found to be missing in NCBI protein. The C-terminal sequence of OsGH3-6 from TIGR did not show any homology to GH3 proteins. After alignment of all the three OsGH3-6 cDNA sequences with corresponding genomic sequence [bacterial artificial chromosome (BAC) clone OJ1607_F09], it was concluded that KOME sequence is in order and was thus included in Table 1 for further analysis.

Genomic organization and chromosomal distribution

A comparison of the full-length cDNA sequences with the corresponding genomic DNA sequences showed that the coding sequences of *OsGH3* genes are disrupted by one to four introns except for *OsGH3-6* and *-9*, which do not

Fig. 1 Exon–intron organization of rice *GH3* genes. The coding and untranslated regions are represented by *black* and *open boxes*, respectively. *Lines* represent the introns. The numbers 0, 1, and 2 represent phase 0, 1, and 2 introns, respectively



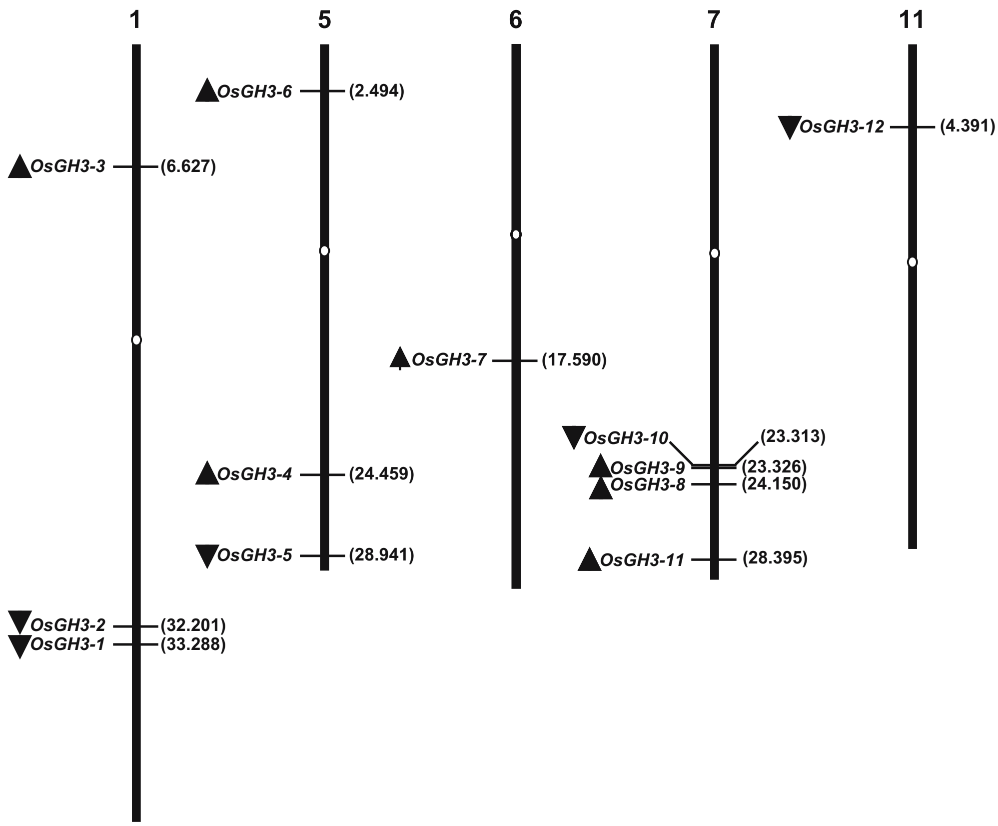


Fig. 2 Genomic distribution of *GH3* genes on rice chromosomes. White ovals on the chromosomes (*vertical bars*) indicate the position of centromeres. The arrows next to gene names show the direction of transcription. The numbers in parentheses designate the position of the first exon of corresponding *GH3* gene in megabases (Mb) on rice chromosome pseudomolecules available at TIGR (release 3). The chromosome numbers are indicated at the top of each bar

harbor any intron (Table 1). Although the exon–intron organization of *OsGH3* genes in terms of their numbers (Fig. 1) does not seem to be very similar, their intron phasing is highly conserved, which is indicative of exon shuffling (Kolkman and Stemmer 2001). The bacterial artificial chromosome (BAC) or phage artificial chromosome (PAC) clones carrying the *OsGH3* genes were identified (Table 1). The chromosome map positions of BAC/PACs given in centiMorgans (cM) from top of the chromosome and the nearest marker to each *OsGH3* gene are indicated in Table 1. Also, the position (in megabases) and direction of transcription (arrows) of each gene was determined on rice chromosome pseudomolecules as shown in Fig. 2.

The 12 rice *GH3* genes are distributed on only five of the 12 rice chromosomes. Four of the *OsGH3* genes are present on chromosome 7, three on chromosomes 5 and 7 each, whereas only one *GH3* gene is present each on chromosomes 6 and 11, respectively (Fig. 2). The distribution of 12 members of this multigene family did not reveal evident clusters. However, in *Arabidopsis*, two clusters of *GH3* genes were found, one on chromosome V containing five genes (*AtGH3-12* to *-16*) and the other on chromosome I containing three genes (*AtGH3-18* to *-20*) organized in tandem in the same orientation (Hagen and Guilfoyle 2002). The amino acid sequences of AtGH3-13 to -16 (74% to 85% identity) and AtGH3-18 and -19 (74% identity) are highly homologous. These results suggested that they arose relatively recently by gene duplication and may have similar or overlapping functions and also account for an enlarged *GH3* gene family in *Arabidopsis* in comparison to rice.

Sequence analysis and phylogenetic relationship

Arabidopsis GH3 proteins did not show the presence of any known conserved motif or domain even within the highly conserved regions. However, they were found to be distantly related (13% sequence identity) to the acyl adenylate-forming firefly luciferase super family (Staswick et al. 2002). OsGH3 proteins also did not show any match in the currently recognized motifs in the databases. The multiple-sequence alignment of the full-length OsGH3 proteins was done using ClustalX (Fig. 3). The core region showed significant conservation (Fig. 3a; marked with arrows) as compared to the N- and C-terminal regions of predicted GH3 proteins. Pairwise analyses with the full-length protein sequences indicate that the overall identities range from 17% to 78% (Table 2). Relatively high identities were found between OsGH3-1 and -4 (ca 78%), OsGH3-8 and -9 (ca 75%). Such high homologies may indicate that they may perform essentially similar functions. However, other family members show a variation of 17% to 61% identity among them, which may indicate their distinct origin and function.

To examine the phylogenetic relationship among the OsGH3 proteins, an unrooted tree was constructed from alignment of their full-length protein sequences (Fig. 4). Two major groups (groups I and II), with very strong boot-

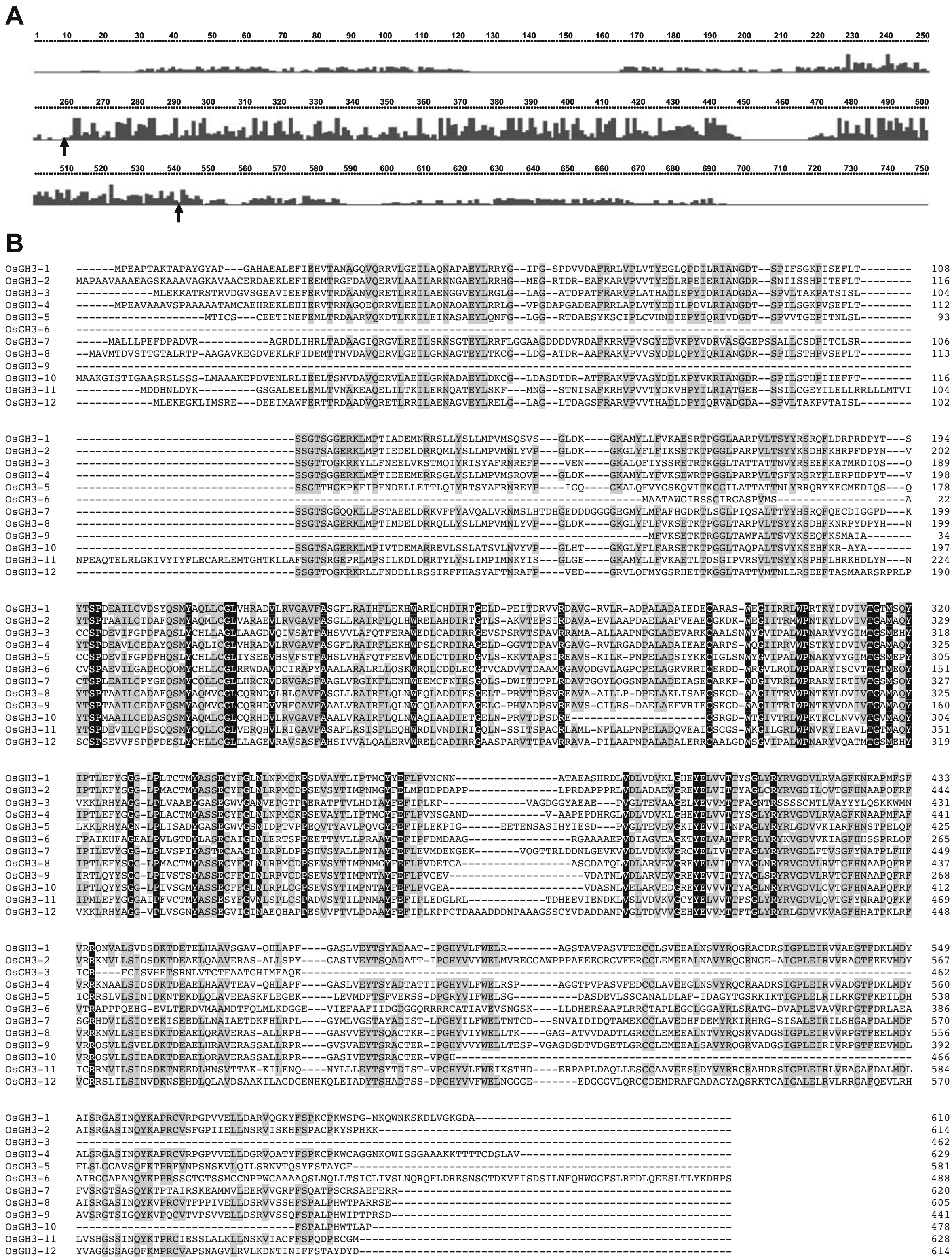


Fig. 3 (**A**) Alignment profile of rice GH3 proteins obtained with the ClustalX program. The height of the bars indicates the number of identical residues per position. The region between the arrows indicates the core region with high-sequence similarity. (**B**) Multiple alignments of the full-length rice GH3 proteins obtained with ClustalX. Fully and partially conserved (present in more than 50% of aligned sequences) residues are highlighted in *black* and *gray boxes*, respectively. Gaps (marked with *dashes*) have been introduced to maximize the alignments

Table 2 Percentage identities among full-length OsGH3 proteins

	OsGH3-1	OsGH3-2	OsGH3-3	OsGH3-4	OsGH3-5	OsGH3-6	OsGH3-7	OsGH3-8	OsGH3-9	OsGH3-10	OsGH3-11	OsGH3-12
OsGH3-1	***	59.3	30.3	**77.7**	31.5	25.6	40.8	60.5	52.6	50.4	44.9	30.3
OsGH3-2		***	30.7	60.3	32.7	25.6	32.2	72.1	61.0	60.5	46.3	34.3
OsGH3-3			***	31.4	52.4	20.8	30.1	30.1	**17.0**	27.7	25.5	58.0
OsGH3-4				***	31.0	25.2	39.4	60.3	52.8	50.6	44.2	31.6
OsGH3-5					***	24.4	24.1	32.9	32.4	29.7	27.7	49.9
OsGH3-6						***	24.4	26.0	26.8	19.0	24.8	25.8
OsGH3-7							***	35.9	40.1	33.3	38.7	29.9
OsGH3-8								***	74.6	69.5	46.3	32.2
OsGH3-9									***	56.0	43.5	32.9
OsGH3-10										***	40.8	30.3
OsGH3-11											***	26.9
OsGH3-12												***

Highest and lowest identities are presented in bold
Tripple asterisk (***) refers to 100 percent identity between the same protein

strap support, were observed, which included four and eight OsGH3 proteins, respectively. Twelve OsGH3 proteins formed only three sister pairs. The sister pair OsGH3-1 and -4 was found to be present on duplicated chromosomal block 1 present between chromosomes 1 and 5 as described by Paterson et al (2004). However, the sister pair OsGH3-9 and -10 may represent a local duplication event. The third sister pair (OsGH3-3 and -12) may also be present on unidentified duplicated chromosomal block. Taken together, these observations throw some light on the diversification of rice *GH3* genes during evolution.

The *Arabidopsis* GH3 proteins have been classified into three groups on the basis of their protein structure and specificity to adenylate plant hormones (Staswick et al 2002). To examine protein relationships of rice and *Arabidopsis* GH3 proteins, an unrooted tree was constructed from alignments of their full-length GH3 protein sequences (Fig. 5). Based upon the sequence homology, all the rice and *Arabidopsis* GH3 proteins clustered distinctly into three groups (I, II, and III). Group I consisted of only six members, four from rice cluster, i.e., OsGH3-3, -5, -6, and -12, along with AtGH-10 and -11. AtGH3-11 (JAR1/FIN219) has been shown to adenylate jasmonic acid (JA) (Staswick et al. 2002). Seven of the OsGH3 proteins belonged to group II AtGH3 proteins, which are involved in IAA adenylation (Staswick et al. 2002). The group II *Arabidopsis GH3* genes encode IAA-amido synthetases that conjugate amino acids to IAA and regulate plant growth and development (Staswick et al. 2005). However, group III included only *Arabidopsis* GH3 proteins with unknown function, which do not adenylate IAA, JA, or SA. Surprisingly, no group III GH3 protein could be identified in rice sequence databases searched. It can be speculated that group III GH3 proteins were lost in rice after divergence of monocots and dicots or evolved in dicots after the divergence from monocots and may perform dicot-specific functions. However, OsGH3-7 grouped with group II rice GH3 proteins (Fig. 4) but did not group with *Arabidopsis* group II GH3 proteins (Fig. 5).

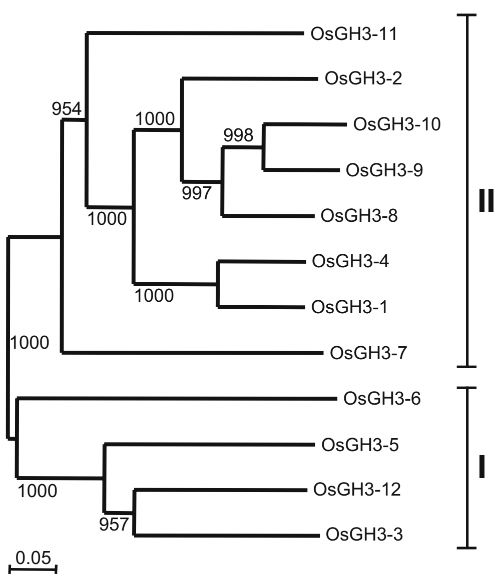


Fig. 4 Phylogenetic relationship among the rice GH3 proteins. The unrooted tree was generated using ClustalX program by neighbor-joining method. Bootstrap values from 1000 replicates are indicated at each node

Differential expression of *OsGH3* genes

The frequency of ESTs or cDNAs available in different databases has been considered as a useful tool for preliminary analysis of gene expression (Adams et al. 1995). Searching different databases, ESTs were identified for most of the *OsGH3* genes, but the frequency of ESTs for individual genes varies greatly (Table 1; Supplementary material, Table S3). For example, eight ESTs are deposited for *OsGH3-2* and 28 for *OsGH3-5*, whereas no ESTs could be identified for *OsGH3-1*, *-6*, *-9*, *-10*, and *-12* (Table 1). The full-length cDNA clones have been identified for *OsGH3-1* through *OsGH3-9*. The transcription of *OsGH3-*

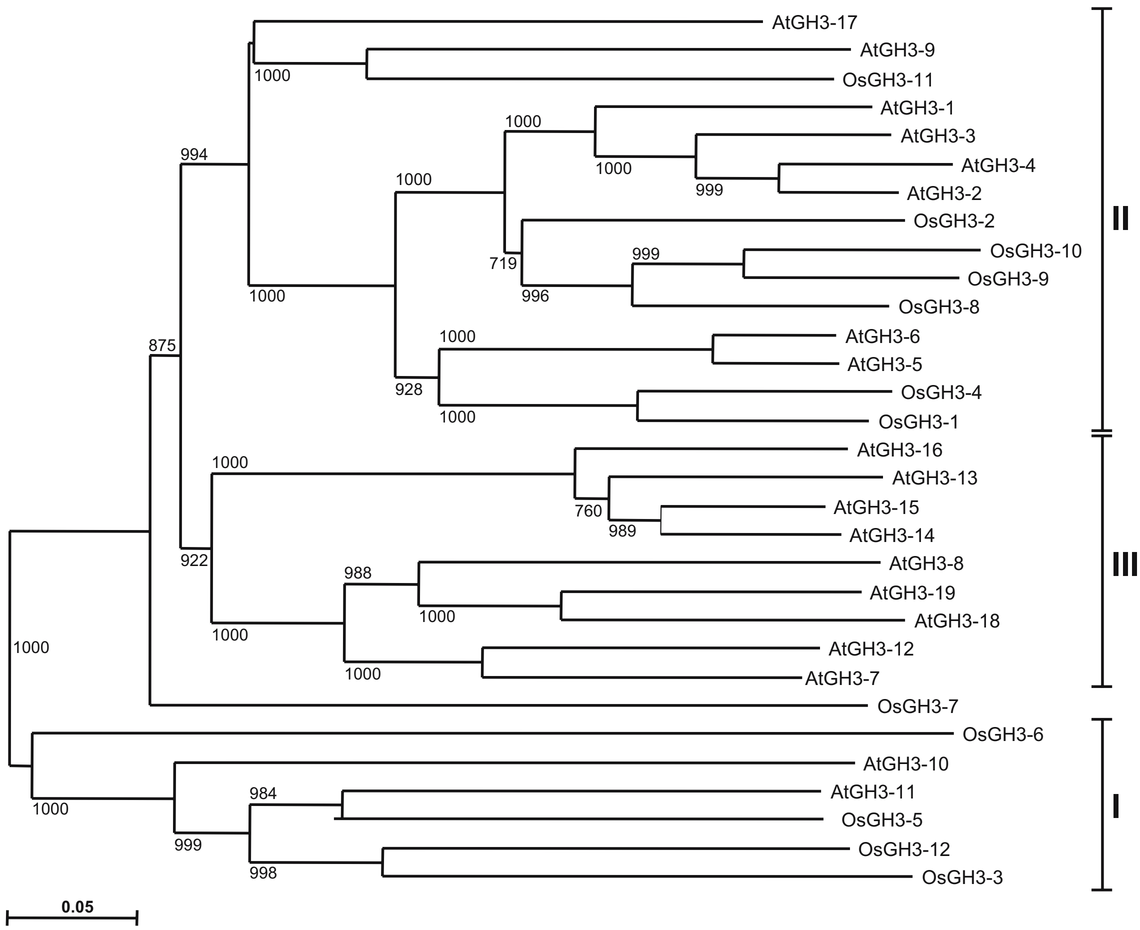


Fig. 5 Phylogenetic relationship of rice and *Arabidopsis* GH3 proteins. The unrooted tree was generated using ClustalX program by neighbor-joining method. Bootstrap values (above 50%) from 1000 replicates are indicated at each node. The AGI gene names of *Arabidopsis* GH3 proteins are: AtGH3-1, At2g14960; AtGH3-2/YDK1, At4g37390; AtGH3-3, At2g23170; AtGH3-4, At1g59500; AtGH3-5, At4g27260; AtGH3-6/DFL1, At5g54510; AtGH3-7, At1g23160; AtGH3-8, At5g51470; AtGH3-9, At2g47750; AtGH3-10, At4g03400; AtGH3-11/FIN219/JAR1, At2g46370; AtGH3-12, At5g13320; AtGH3-13, At5g13350; AtGH3-14, At5g13360; AtGH3-15, At5g13370; AtGH3-16, At5g13380; AtGH3-17, At1g28130; AtGH3-18, At1g48660; AtGH3-19, At1g48670. AtGH3-20 has not been included as it codes for a partial GH3 protein

10 and *-12* genes, however, remained unclear, as no corresponding full-length cDNA or EST could be identified from the databases. The expression of all the 12 *OsGH3* genes was thus verified experimentally and quantitated by real-time PCR analysis in the present study.

Many of the *GH3* genes in *Arabidopsis*, soybean, and tobacco were found to be differentially expressed in different tissues or in response to exogenous auxin and light stimuli (Tepperman et al. 2001; Hagen and Guilfoyle 2002; Tanaka et al. 2002; Takase et al. 2004). To determine the organ-specific expression pattern of each *OsGH3* gene, real-time PCR was performed with total RNA isolated from etiolated shoots, green shoots, roots, flowers, and callus. This analysis revealed that *OsGH3* genes are differentially expressed in various tissues/organs (Fig. 6a). The transcription of *OsGH3-12* was found to be root-specific and was hardly detectable in other tissues examined (Fig. 6a). However, *OsGH3-10* was found to be expressed at very low levels in all the tissues. In contrast, other *OsGH3* genes were found to be expressed in almost all the tissues examined but at different levels. Also, significant differences were found in the transcript abundance of *OsGH3* genes in etiolated and green shoots (Fig. 6a), indicating their light regulation. The transcript levels of most of the *OsGH3* genes were upregulated by auxin treatment as is evident from the data presented in Fig. 6b, although to varying degree. The effect is more pronounced on *OsGH3-1*, *-2*, *-4* and, to some extent, on *OsGH3-8* too. The difference in the inducibility of individual *GH3* genes is likely due to a variety of factors such as kinetics of induction, tissue-specific auxin reception, cell type-dependent and differential regulation of free auxin concentrations, or different modes of auxin-dependent transcriptional and posttranscriptional regulation.

EST analysis in other monocots

A search of Genbank database showed that *GH3*-related genes are present throughout the plant kingdom. ESTs have been found for GH3 proteins in blue-green alga, *Synechocystis*, and dicots (*Arabidopsis*, tobacco, soybean, *Capsicum*). An extensive analysis of the EST sequences in

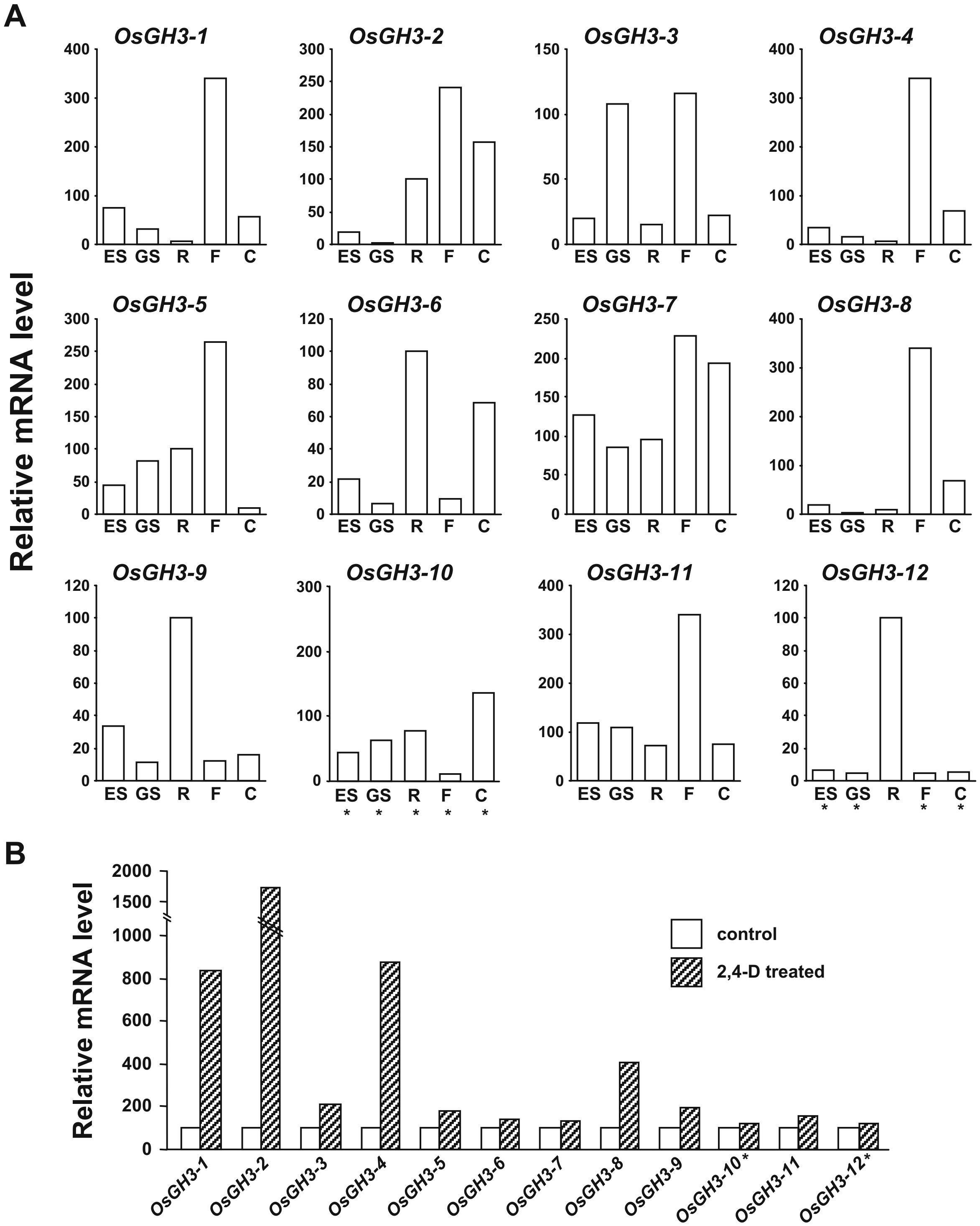


Fig. 6 Real-time PCR expression profiles of individual *OsGH3* genes. (**A**) The relative mRNA levels of individual *OsGH3* genes normalized with respect to housekeeping gene *UBQ5* in different tissues (*ES* etiolated shoots, *GS* green shoots, *R* roots, *F* flowers, *C* callus). (**B**) The relative mRNA levels of individual *OsGH3* genes in control (16 h auxin-depleted) and 2,4-D (30 μm)-treated coleoptile segments of 3-day-old etiolated rice seedlings. *Asterisks* indicate that the expression was close to the detection limit

Genbank database revealed the presence of a large number of ESTs in other monocots like *T. aestivum*, *Z. mays*, *S. bicolor*, *S. officinarum*, and *H. vulgare* with high-sequence similarity with *OsGH3* genes (Supplementary material, Table S3). These results suggest that the GH3 proteins may be present in diverse plant species and are likely to play important roles in plant cellular processes. However, like in case of *OsGH3* gene family, no ESTs corresponding to group III *GH3* genes of dicots could be identified in other monocots too. What precisely is the evolutionary significance of this observation will only be unraveled through functional validation of this group of genes in dicots like *Arabidopsis*.

What are the functions of OsGH3 proteins?

The GH3 proteins in *Arabidopsis* have been shown to be involved in plant growth and development, photomorphogenesis, light- and auxin-signaling, and auxin homeostasis (Hsieh et al. 2000; Nakazawa et al. 2001; Tepperman et al. 2001; Tanaka et al. 2002; Takase et al. 2004; Staswick et al. 2005). The results of structural analyses of rice GH3 proteins will be helpful in their functional analysis. Our analysis showed that GH3 proteins from *Arabidopsis* and rice are divided into three and two groups, respectively. GH3 proteins classified in the same groups may have the similar functions in events common to both monocot and dicot plants. Group III GH3 proteins may have dicot-specific functions as they could not be identified from rice or other monocots. Organ-specific differential expression suggests diverse roles of these proteins during plant growth and development. The effect of light and auxin treatment on the transcript levels of *OsGH3* genes implies their role in light and auxin signal transduction.

To further elucidate the functions of these genes, we investigated the phenotypes of rice *Tos17* retrotransposon insertion mutants (Miyao et al. 2003) of these genes with the aid of *Tos17* mutant panel database (http://www.tos.nias.affrc.go.jp/) using the BLAST program. We could enlist eight insertion mutants corresponding to two (*OsGH3-5* and *-7*) of the 12 *OsGH3* genes (Supplementary material, Table S4). The phenotypes of the insertion mutants of these genes showed dwarfism, sterility, vivipary, and leaf withering. From these phenotypes, it can be speculated that these genes may play a role in different metabolic pathways and cellular processes influenced by light and auxin, although the assumption that these phenotype changes are specific to insertional mutagenesis of *OsGH3* gene remains to be proven experimentally. A detailed analysis of these insertional mutants already available and RNAi strategy for the remaining *OsGH3* genes will greatly help in elucidation of the role of these genes and their functional validation.

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